PROJECT BRIEF · FOR THE INTERN

AI Business Growth & Trend Intelligence Platform

The features, how they connect, the AI that powers them, and how to build the advanced parts — a growth analyst, strategy & recommendations, video-hook analysis, agents, trend prediction, data access, and competitor discovery.

Built by the intern | Core + advanced features | You choose the models

1 The idea, in one line

> A tool that shows a business **what content is working in its market**, and helps them **create better content, faster.**

The app looks at a business's competitors and current trends, then hands the owner a plan and ready-to-post content ideas. It's **six features** connected in **one simple loop**, with a set of **advanced capabilities** layered on top.

In engineering terms: a modular AI SaaS platform with a **data → analysis → strategy → content** pipeline, powered by LLMs + analytics + optional multimodal extensions.

2 What the app does — six features

1 Business Onboarding

A short form for the basics: industry, country, product or service, target audience, goals — and the user's known **competitors**.

Output: a **business profile** + a starter competitor list that everything else builds on.

Starts simple: a form, not a deep AI interview.

2 Competitor Tracker

Pulls your competitors' latest posts, engagement (likes / comments), and posting frequency. This content is the **raw material the trend engine reads**.

Output: each competitor's **top-performing content**, ready to learn from.

Note: add them at onboarding or here; auto-discovery & how to access their content are in section 6.

3 Trend Detection

Works out what's trending **from your competitors' content plus the wider niche** — the recurring topics, formats, and hooks that are actually performing. Grounded in real content, not a vague feed.

Output: trending **topics, formats, and hooks** for your niche this week.

Pluggable sources: start with one, add more later (competitors, hashtags, datasets) without a rewrite. More in section 6.

4 Content Generator MAIN FEATURE

Takes the profile + competitor insights + trends and writes the content — then **scores and ranks** the ideas so the best rise to the top. The payoff of the whole app.

Output: 10 post ideas (captions, hooks, hashtags, CTAs), each with an **engagement-potential** and **trend-match** score — so it's clear why these ranked top.

Note: text only — powered by agents (section 6).

5 Strategy & Recommendations THE DECISION LAYER

Sits between the insights and the writing: it turns everything the app has learned into a **plan**, and tells the owner exactly **what to do next**. This is the strategy brain the content generator then acts on.

Output: a recommended **strategy** — posting calendar, best times to post, best hashtags, and target audience — plus a **prioritised list of next-best moves** (which trend to ride, which content to make next), each with a reason.

How it works: mostly the LLM synthesising the profile + competitors + trends, a simple **ranking score** (trend momentum + fit to the business + past feedback) to order the suggestions, and engagement math for best time & hashtags. It gets sharper as the owner gives feedback.

6 Business Analyst YOUR GROWTH AUDIT

Acts like an **analyst reviewing the business**: what it's doing well, where it's weak versus the market and competitors, and what to change to **reach more people**. This is the "strengths, weaknesses & what to improve" view the owner asked for.

Output: a **SWOT-style read** (strengths, weaknesses, opportunities, threats) plus a short list of concrete **growth moves** — each tied to a signal, e.g. *"competitors post 2× more video; your reach is capped by low posting frequency."*

How it works: an **Analyst** compares your business profile against competitor data + trends to spot gaps and edges; a **Strategist** turns those into prioritised growth actions. Grounded in data the app already collects, it can run as two agents and refreshes as new data arrives.

3 The core loop

This is the whole system. Everything serves this one flow:

1. **Onboard the business**
   Fill the form & add your known competitors → save the profile.
2. **Collect content**
   Pull your competitors' posts + wider niche content (via the data provider).
3. **Understand trends from it**
   From that content, surface the topics, formats & hooks that are performing.
4. **Analyse, score & strategise**
   Read strengths & weaknesses, score the options, recommend a plan and the next-best moves.
5. **Generate ranked content**
   Get 10 ideas, each ranked by predicted engagement & trend-match.
6. **Owner posts manually**
   The person reviews and publishes themselves — the app never auto-posts.
7. **Feedback (optional)**
   The owner marks what they liked → scores & recommendations get sharper next time.

4 The six screens

Dashboard

A one-glance overview: the business, this week's trends, tracked competitors, and fresh content ideas.

Competitors

The competitor list, each one's posts, and their engagement numbers.

Trends

Trending topics, fast-growing hashtags, and the viral posts behind them.

Content Generator

A "Generate ideas" button that returns captions, hooks, hashtags, and CTAs.

Strategy & Recommendations

The recommended plan — posting calendar, best times, hashtags, audience — and a ranked list of the next-best content and trends to act on, each with a reason.

Business Analysis (SWOT & Growth)

Your strengths and weaknesses versus the market, and a short list of concrete moves to grow reach — each linked to the signal behind it.

5 The models that power the app

You don't train anything from scratch. You call existing models through an API and write the logic around them. For the core app, three things do the work:

Text-generation model (LLM) CORE · THE ENGINE

Turns the business profile + competitor insights + trends into the post ideas, captions, hooks, hashtags, and CTAs. Powers the main feature.

Text analysis for trends READS THE POSTS

Reads competitor and social posts, groups them into topics, and pulls out recurring hooks and formats. The same LLM can do this, or simple keyword/hashtag logic.

Engagement math JUST LOGIC, NO ML

Plain statistics, not a trained model: rank posts by engagement, flag hashtags growing fast, and mark posts whose engagement spikes as "viral".

Choosing the models is your call

Which specific LLM, multimodal model, or embeddings model to use is left to you. Compare current benchmarks for quality, speed, and cost, and pick what fits the budget — the app is built so any provider can be swapped in behind an API.

6 Advanced features — how they work

The capabilities layered on top of the core loop. Each note shows a rough build difficulty. Build the thin version first, then improve.

1 Video-hook analysis (in Trend Detection)

MEDIUM

What: given a video URL, find where the catchy part — the hook — is, and explain why it works.

How: pull the video (a downloader or the data provider) → transcribe it with timestamps (speech-to-text) → mark "moments" with scene-cut, audio-peak, and on-screen-text detection → send the transcript + a few sampled frames to a **multimodal model** to locate the hook (in short-form it's almost always the first 1–3 seconds) and explain why. Feed the result into the content generator as a hook style to copy.

Optional for the first version — it leans heavily on data availability, so treat it as an enhancement, not a launch blocker. Limit: without the platform's real retention data you're reading the content, not viewer behaviour — good enough for short-form. Active research area ("highlight detection" / a video's "hooking period").

2 Agentic content generation

LOW–MEDIUM

What: replace one big prompt with a small team of agents that research, strategise, write, and self-edit.

How: give each agent a role — **Researcher** (gathers profile + competitors + trends), **Strategist** (picks angle & format), **Writer** (drafts), **Editor** (checks brand voice and sends weak drafts back). An agent framework orchestrates them; the human approves the final output.

Options: a role-based framework (fast to prototype) or a graph-based one (more control and human-in-the-loop). Pick after a quick comparison — both are model-agnostic.

3 The data / "scraping" solution

LOW TO WIRE

What: get competitor and trend data reliably, without building a fragile scraper.

How: don't reverse-engineer platforms yourself — it breaks constantly and violates their terms. Use a **third-party social-data provider** that returns public data as clean JSON, or start from a downloaded dataset. Begin with one platform, then add more.

Caveat: providers pull public data (courts have generally allowed that), but it's a terms-of-service grey area — keep it visible. Official APIs (TikTok's Research API, Meta's Graph API) are gated or limited, so most products use a provider. Main cost is the subscription.

4 Trend prediction engine

LOW–MEDIUM

What: not just what's trending now, but what's growing and how long it may last.

How: treat each topic / hashtag / sound as a daily **time series** → compute momentum (is the slope positive and accelerating?) to label it **emerging · peaking · fading** → add a simple forecast (moving average or exponential smoothing) for a growth estimate. A trained classifier is a later upgrade, not the starting point.

Present it as a confidence signal, not a certainty.

5 Automatic competitor discovery

MEDIUM

What: automatically find other businesses **similar to yours**, so the user doesn't have to add every competitor by hand.

How: take the business profile as a fingerprint → find candidates via search + the data API's keyword / similar-account endpoints → rank how close each one is using an **embeddings model** (cosine similarity of what they sell / post) plus shared-hashtag overlap → show the closest matches for the user to confirm. Store the vectors inside the existing database (a Postgres vector extension) — no separate vector database needed.

Suggest, don't auto-add — the user confirms.

7 Mini glossary

Business profile
The saved facts about a business (industry, product, audience, goals) that guide the AI.

Multimodal model
An AI that understands more than text — here, video frames + audio + transcript together.

Agent / crew
Specialised AI helpers (researcher, writer, editor) that work together under a coordinator.

Momentum
How fast a trend is growing — used to call it emerging, peaking, or fading.

Recommendation engine
The part that ranks and suggests the next-best actions, each with a reason.

SWOT
A quick read of Strengths, Weaknesses, Opportunities, and Threats for a business.

Content scoring
Ranking each idea by an engagement-potential and trend-match score, so the best surface first.

LLM
The AI model that reads and writes text — used through an API.

Embeddings
Turning text into number-vectors so you can measure how similar two things are.

Hook
The opening line or first seconds of a video that grab attention.

Engagement
Likes + comments + shares — how much a post resonates.

Posting calendar
A schedule of what to post and when, produced as part of the strategy.

Reach
How many people see the content — the thing the growth moves aim to increase.

Pluggable source
A data source added behind one interface — you can add more without rewriting the app.